AI calcs

same colour = same number · yellow = worked out · red = careful

L2–3 SEARCH

A* vs UCS vs Greedy (which node next)

P: g 120, h 200 Q: g 90, h 250 R: g 150, h 160

$f = \underset{g}{\text{cost so far}} + \underset{h}{\text{guess to goal}}$

A*: P 120+200 = 320, Q 90+250 = 340, R 150+160 = 310
→ smallest f → **R (310)**

UCS: smallest g only → **Q (90)** g = add up road costs so far

Greedy: smallest h only → **R (160)**

A* is optimal only if h never overestimates (admissible)

Simulated annealing accept prob

ΔE = 3 (worse by 3) T = 2

$P = e^{-\Delta E / T} = e^{-3/2} = e^{-1.5}$

$= 0.2231$

better move → always accept. high T = accepts more bad moves

GA roulette wheel selection

fitness 24 30 16 8

$P(\text{pick}) = \frac{\text{fitness}}{\text{sum of all fitness}}$

sum = 24 + 30 + 16 + 8 = 78

$P = \frac{24}{78} = 0.3077$

N-queens successors

successors = n queens × (n − 1) other rows

$= 8 \times 7 = 56$

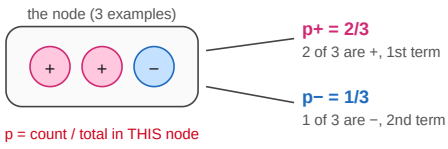
4-queens → 4 × 3 = **12**

each queen can go to every row except the one it is on

L4 LEARNING FROM EXAMPLES

Entropy & Gini of a node

2 + 1 −



p = count / total in THIS node

$H = -p_+ \log_2 p_+ - p_- \log_2 p_-$

$= -\frac{2}{3}\log_2\frac{2}{3} - \frac{1}{3}\log_2\frac{1}{3}$

$= 0.39 + 0.5283 = 0.9183$

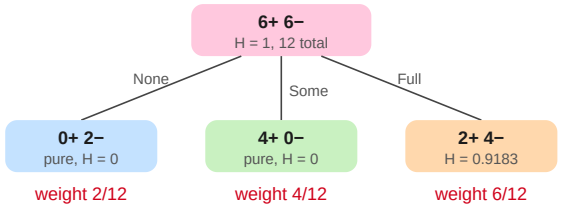
$Gini = 1 - (p_+^2 + p_-^2)$

$= 1 - (\frac{4}{9} + \frac{1}{9}) = 0.4444$

pure node → 0. 50/50 → H = 1, Gini = 0.5

Information gain (restaurant, Patrons)

root 6+ 6− None 0+ 2− Some 4+ 0− Full 2+ 4−



$IG = H(\text{root}) - \Sigma \frac{\text{branch size}}{\text{total}} H(\text{branch})$

H(root) = 1 None = 0, Some = 0 pure

H(Full) = H(2/6, 4/6) = 0.9183

$IG = 1 - [\frac{2}{12}(0) + \frac{4}{12}(0) + \frac{6}{12}(0.9183)]$

$= 1 - 0.4591 = 0.5409$

Gini: root 0.5, Full 0.4444 → $0.5 - \frac{6}{12}(0.4444) = 0.2778$

pick the attribute with the BIGGEST gain. every branch 50/50 (Type) → IG = 0

Sum of squared errors

h(x) = 1 + 2x (1,4) (2,4) (3,8)

$SSE = \Sigma (\underset{y}{\text{actual}} - \underset{h(x)}{\text{predicted}})^2$

h(1) = 3, h(2) = 5, h(3) = 7

$= (4-3)^2 + (4-5)^2 + (8-7)^2$

$= 1 + 1 + 1 = 3$

MSE → divide by n (= 1). some slides use ½ SSE

KNN classify (k = 3)

query (2,3) A (1,1) B (3,4) B (4,3) A (0,5) A (6,6)

$d = \sqrt{(x_1-x_2)^2 + (y_1-y_2)^2}$

A(1,1): √(1+4) = 2.24 B(3,4): √(1+1) = 1.41

B(4,3): √(4+0) = 2 A(0,5): √(4+4) = 2.83

A(6,6): √(16+9) = 5

3 closest: B, B, A → **B**

no need to square root if you only compare

Precision, recall, F1

TP 40 FP 10 FN 20 TN 30

$acc = \frac{TP + TN}{all}$ $P = \frac{TP}{TP + FP}$ $R = \frac{TP}{TP + FN}$

$acc = \frac{40 + 30}{100} = 0.7$

$P = \frac{40}{40 + 10} = 0.8$ $R = \frac{40}{40 + 20} = 0.667$

$F1 = \frac{2PR}{P + R}$

$= \frac{2(0.8)(0.667)}{0.8 + 0.667} = 0.7273$

P → of the ones I said yes. R → of the real yes

L5 NEURAL NETS

Perceptron output + update

x = (1, 1) w = (0.3, 0.2) b = −0.4 target y = 0 η = 0.1

out = step($w \cdot x + b$)

= 0.3(1) + 0.2(1) − 0.4 = 0.1 ≥ 0 → out = 1

w_1 new = $w1 + \eta (y - \text{out}) x1$

= 0.3 + 0.1(0 − 1)(1) = **0.2**

y = target, out is 0 or 1 (not 0.1). right answer → no change

Forward pass (ReLU then sigmoid)

x1 = 0 x2 = 2 w1,3 = −0.2 w2,3 = −0.4 w3,5 = 1.3

$z_3 = x1\,w1{,}3 + x2\,w2{,}3$

= 0(−0.2) + 2(−0.4) = −0.8

a_3 = ReLU(−0.8) = 0

$z_5 = 0 \times 1.3 = 0 \rightarrow \sigma(0) = \frac{1}{1+e^{0}} =$ **0.5**

ReLU: negative → 0. $\sigma(z) = 1 / (1 + e^{-z})$

Activations & loss (quick ones)

$\sigma(z) = \frac{1}{1+e^{-z}}$ ReLU(z) = max(0, z) tanh(0) = 0

$\sigma(0) = \frac{1}{1+1} =$ **0.5** ReLU(−3) = **0**

loss = $(y - \hat{y})^2$

= $(1 - 0.8)^2 = 0.2^2 =$ **0.04**

square AFTER subtracting

L6 REINFORCEMENT LEARNING

Discounted return

rewards −1, −1, +10 γ = 0.5

$G = r_1 + \gamma r_2 + \gamma^2 r_3$

= −1 + 0.5(−1) + 0.25(10)

= −1 − 0.5 + 2.5 = **1**

first reward is NOT discounted

Q-value with slippery moves

0.8 intended 0.1 / 0.1 sideways R = −0.04 γ = 0.5

V: 0.39, 0.25, 0.04

Q = Σ P($R + \gamma V(s')$)

= 0.8(−0.04 + 0.5×0.39) + 0.1(−0.04 + 0.5×0.25) + 0.1(−0.04 + 0.5×0.04)

= 0.8(0.155) + 0.1(0.085) + 0.1(−0.02)

= 0.124 + 0.0085 − 0.002 = **0.1305**

bump into wall → you stay, use your own V

Pick the best action (policy)

A = 0.32 B = 0.44 C = 0.21

value(move) = 0.8 × target + 0.1 × left side + 0.1 × right side

Up = 0.8(0.32) + 0.1(0.44) + 0.1(0.21) = 0.321

Right = 0.8(0.44) + 0.1(0.32) + 0.1(0.21) = 0.405

biggest → **Right**

policy = argmax, so give the ACTION not the number

TD & Q-learning

V(s) = 0.52 V(s') = 0.21 α = 0.2 γ = 0.9 r = 0

$V(s) \leftarrow V(s) + \alpha (r + \gamma V(s') - V(s))$

= 0.52 + 0.2(0 + 0.9×0.21 − 0.52)

= 0.52 + 0.2(−0.331) = **0.4538**

Q(s,a) = 0.97 max Q(s') = 1.63

$Q \leftarrow Q + \alpha(r + \gamma \max Q(s',a') - Q)$

= 0.97 + 0.2(0 + 0.9×1.63 − 0.97)

= 0.97 + 0.2(0.497) = **1.0694**

Q-learning uses the MAX of next state. SARSA uses the action taken. U(s) = V(s)

Expected utility (MEU)

EU = Σ P × U

A = 0.8(10) + 0.2(−10) = 8 − 2 = 6

B = 1(7) = 7

7 > 6 → **B**

MEU = biggest expected value, even if A can win more

Direct utility estimate

U(s) = average of reward-to-go seen from s

= $\frac{0.84 + 0.92}{2} =$ **0.88**

reward-to-go = sum of rewards from that state to the end

Q(s,a) from a Q-table + ε-greedy

Up 0.4 Down 0.7 Left −0.2 Right 0.1 ε = 0.2

$V(s) = \max_a Q(s,a)$ $\pi(s) = \text{argmax}_a Q(s,a)$

max(0.4, 0.7, −0.2, 0.1) = **0.7** → **Down**

P(greedy) = $(1 - \varepsilon) + \frac{\varepsilon}{|A|}$

= 0.8 + $\frac{0.2}{4}$ = 0.8 + 0.05 = **0.85** P(other) = **0.05**

max = value, argmax = action. random pick excludes greedy → 1 − ε and ε/3

SARSA vs Q-learning

Q = 0.5 r = 1 γ = 0.9 α = 0.1 taken 0.2, best 0.8

$Q \leftarrow Q + \alpha (r + \gamma Q(\text{next}) - Q)$

SARSA (next = action TAKEN, 0.2):

= 0.5 + 0.1(1 + 0.9×0.2 − 0.5) = 0.5 + 0.1(0.68) = **0.568**

Q-learning (next = MAX, 0.8):

= 0.5 + 0.1(1 + 0.9×0.8 − 0.5) = 0.5 + 0.1(1.22) = **0.622**

SARSA = on-policy. Q-learning = off-policy. both model-free

Q-learning over episodes

α = 0.5 γ = 1 end reward 10

$Q \leftarrow Q + \alpha (r + \gamma \max Q(\text{next}) - Q)$ terminal → max Q(next) = 0

ep 1, A: 0 + 0.5(0 + 1×0 − 0) = 0

ep 1, B: 0 + 0.5(10 − 0) = 5

ep 2, A: 0 + 0.5(0 + 1×5 − 0) = **2.5**

update in the order visited. the reward reaches A one episode later

Q(s,a): deterministic or terminal

$Q(s,a) = \overset{-1}{r} + \overset{0.9}{\gamma} \max \overset{5}{Q(s',a')}$

$= -1 + 0.9 \times 5 = \underline{\mathbf{3.5}}$

terminal: target = $\overset{5}{r}$ only

$Q = 2 + 0.5(5 - 2) = \underline{\mathbf{3.5}}$

no γ × anything after a terminal state

L7 CNN

Conv output size & parameter count

input 7×7×16 32 filters filter 5×5 stride 2 padding 2

$$\text{output} = \frac{\overset{7}{\text{input}} + 2\,\overset{2}{\text{padding}} - \overset{5}{\text{filter size}}}{\underset{2}{\text{stride}}} + 1$$

$= \frac{7 + 4 - 5}{2} + 1 = 3 + 1 = 4 \rightarrow \underline{\mathbf{4 \times 4 \times 32}}$

params = ($\overset{5}{\text{filter}} \times \overset{5}{\text{filter}}$) × $\overset{16}{\text{channel}}$ × $\overset{32}{\text{conv layer}}$

$= 5 \times 5 \times 16 \times 32 = \underline{\mathbf{12\,800}}$

+ bias → + 1 per filter = 12 800 + 32 = 12 832

channel = depth of the INPUT (RGB = 3). depth of output = no. of filters

Conv, one output value

3	1	0
2	0	1
1	4	1

∗

1	0	−1
1	0	−1
1	0	−1

value = sum of (patch × kernel) + bias

$= (3-0) + (2-1) + (1-1) + 0$

$= 3 + 1 + 0 = \underline{\mathbf{4}}$

middle column × 0 drops out. slide by stride for the next one

Max & average pooling (2×2, stride 2)

5	9	2	3
1	9	3	0
9	6	5	6
4	6	6	0

max: [5 9 1 9] = 9 [2 3 3 0] = 3

[9 6 4 6] = 9 [5 6 6 0] = 6

avg (bottom right): $\frac{5 + 6 + 6 + 0}{4} = \underline{\mathbf{4.25}}$

split into blocks, one number per block → 4×4 becomes 2×2

Receptive field

conv 3 conv 3 pool 2, stride 2 conv 3

r = r + (k − 1) × jump jump ×= stride

start r = 1, jump = 1

conv 3: 1 + 2(1) = 3

conv 3: 3 + 2(1) = 5

pool 2 s2: 5 + 1(1) = 6 → jump = 2

conv 3: 6 + 2(2) = **10**

jump only changes AFTER the strided layer

IoU & NMS

box A (0,0)–(6,4) box B (2,0)–(8,4) threshold 0.5

$$\text{IoU} = \frac{\text{overlap}}{\text{area A} + \text{area B} - \text{overlap}}$$

overlap = (6 − 2)(4) = 16, each box = 24

$\text{IoU} = \frac{16}{24 + 24 - 16} = \frac{16}{32} = \underline{\mathbf{0.5}}$

≥ 0.5 → same object → drop the lower score box

NMS: keep top score, delete boxes overlapping it, repeat

L8 RNN, ATTENTION, NLP

Simple RNN hidden state update

x = 0 w = 0.5 h_{t-1} = 0.1 u = 0.75 b = 0.1

$h_t = \tanh(\overset{0.5\,0}{w\,x} + \overset{0.75\;\;0.1}{u\;\;h_{t-1}} + \overset{0.1}{b})$

$= \tanh(0.5(0) + 0.75(0.1) + 0.1)$

$= \tanh(0.175)$

$= \underline{\mathbf{0.1732}}$

u = recurrent weight (on old h). small input → tanh(z) ≈ z

Softmax attention weights

score 1 = 2 score 2 = 1

$\alpha_1 = \frac{e^{\text{score 1}}}{e^{\text{score 1}} + e^{\text{score 2}}}$

$= \frac{e^{2}}{e^{2} + e^{1}} = \frac{7.3891}{7.3891 + 2.7183}$

$= \frac{7.3891}{10.1073} = \underline{\mathbf{0.7311}}$

$\alpha_2 = 1 - 0.7311 = 0.2689$

equal scores → equal weights (2, 2 → 0.5 each)

Attention output (weighted sum)

α = 0.7, 0.3 v1 = [0, 2] v2 = [2, 0]

out = $\overset{0.7}{\alpha 1}\,v1 + \overset{0.3}{\alpha 2}\,v2$

$= 0.7[0, 2] + 0.3[2, 0]$

$= [0, 1.4] + [0.6, 0] = \underline{\mathbf{[0.6, 1.4]}}$

weights must add to 1 (from softmax)

Bigram probability

count("the cat") = 3 count("the") = 5

$$P(\text{cat} \mid \text{the}) = \frac{\overset{3}{\text{count(the cat)}}}{\underset{5}{\text{count(the)}}}$$

$= \frac{3}{5} = \underline{\mathbf{0.6}}$

sentence prob → multiply each bigram

L10 LOGIC

Logic, evaluate a sentence

P = T Q = F R = F

¬P ∨ Q ∧ R ⇒ R

order: ¬ first, then ∧, then ∨, then ⇒

(¬T) ∨ (F ∧ F) ⇒ F

= (F ∨ F) ⇒ F

= F ⇒ F = **True**

⇒ is only false when T ⇒ F

L11 PROBABILITY & BAYES NETS

Probability rules

P(A) = 0.3 P(B) = 0.5 P(A∧B) = 0.15

P(A∨B) = P(A) + P(B) − P(A∧B)

= 0.3 + 0.5 − 0.15 = **0.65**

P(A∧B) = 0.12 P(B) = 0.4

$$P(A|B) = \frac{P(A \wedge B)}{P(B)}$$

$$= \frac{0.12}{0.4} = \mathbf{0.3}$$

P(¬A) → 1 − P(A)

Joint table, P(a|b)

	b	¬b
a	0.132	0.418
¬a	0.25	0.2

$$P(a|b) = \frac{P(a, b)}{P(b)}$$

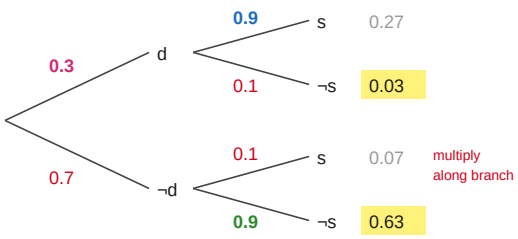
P(b) = P(a,b) + P(¬a,b) = 0.132 + 0.25 = 0.382 ← a cancels out

$$P(a|b) = \frac{0.132}{0.382} = \mathbf{0.3455}$$

to get P(b) add the whole b column

Bayes with a negative result (tree)

P(d) = 0.3 P(s|d) = 0.9 P(¬s|¬d) = 0.9



$$P(d|\neg s) = \frac{P(\neg s|d)\,P(d)}{P(\neg s)}$$

P(¬s) = (1−0.9)(0.3) + 0.9(1−0.3)

= 0.03 + 0.63 = 0.66

$$P(d|\neg s) = \frac{0.03}{0.66} = \mathbf{0.0455}$$

negative result → use the ¬s branches. 1 − P to flip

Disease test (Bayes)

P(D) = 0.01 P(+|D) = 0.9 P(+|¬D) = 0.05

$$P(D|+) = \frac{P(+|D)\,P(D)}{P(+)}$$

P(+) = 0.9×0.01 + 0.05×(1−0.01)

= 0.009 + 0.0495 = 0.0585

$$P(D|+) = \frac{0.009}{0.0585} = \mathbf{0.1538}$$

false alarm rate uses ¬D, which is 0.99 of people

Which machine made the defect

M1 50%, 2% bad M2 30%, 3% bad M3 20%, 5% bad

P(D) = sum of P(M) P(D|M)

= 0.5×0.02 + 0.3×0.03 + 0.2×0.05

= 0.01 + 0.009 + 0.01 = 0.029

$$P(M3|D) = \frac{0.01}{0.029} = \mathbf{0.3448}$$

Bayes net, one full assignment

P(j, ¬m, ¬a, ¬b, ¬e) = P(¬b) P(¬e) P(¬a|¬b,¬e) P(j|¬a) P(¬m|¬a)

each node given its parents only

= 0.999 × 0.998 × 0.999 × 0.05 × 0.99

= **0.0493**

table gives P(a|...) → P(¬a|...) = 1 − that

Bayes net enumeration P(B | j, m)

P(b) = 0.01 P(e) = 0.02 P(a|b,e) .95 .94 .3 .05 P(j|a) .9 / .05

P(m|a) .6 / .02

$$P(b|j,m) = \alpha\, P(b) \sum_e P(e) \sum_a P(a|b,e)\, P(j|a)\, P(m|a)$$

1) j, m part for a and ¬a 2) inside bracket for e and ¬e 3) S(b), S(¬b) 4) divide

P(j|a) P(m|a)

= .9 × .6 = .54

P(j|¬a) P(m|¬a)

= .05 × .02 = .001

S(b)

[b, e] = P(a|b,e)(.54) + P(¬a|b,e)(.001)

= .95(.54) + .05(.001) = 0.51305 ¬a = 1 − .95

[b, ¬e] = P(a|b,¬e)(.54) + P(¬a|b,¬e)(.001)

= .94(.54) + .06(.001) = 0.50766

S(b) = P(b) [P(e) [b, e] + P(¬e) [b, ¬e]]

= .01 [.02(0.51305) + .98(0.50766)]

= .01 [0.010261 + 0.497507] = 0.005078

S(¬b)

[¬b, e] = P(a|¬b,e)(.54) + P(¬a|¬b,e)(.001)

= .3(.54) + .7(.001) = 0.1627

[¬b, ¬e] = P(a|¬b,¬e)(.54) + P(¬a|¬b,¬e)(.001)

= .05(.54) + .95(.001) = 0.02795

S(¬b) = P(¬b) [P(e) [¬b, e] + P(¬e) [¬b, ¬e]]

= .99 [.02(0.1627) + .98(0.02795)] P(¬b) = 1 − .01

= .99 [0.003254 + 0.027391] = 0.030339

$$P(b|j,m) = \frac{S(b)}{S(b) + S(\neg b)}$$

$$= \frac{0.005078}{0.005078 + 0.030339} = \mathbf{0.1434}$$

do b and ¬b the same way, then divide by the total (that is α)

Naive Bayes (text)

"great" in + : 8 words in + : 25 vocab : 10 "great" in − : 0

words in − : 14 P(+) = P(−) = 0.5

$$P(\text{great}|+) = \frac{\text{word appears} + 1}{\text{words in +} + \text{vocab}} = \frac{8}{25} \frac{+1}{+10}$$

$$= \frac{8 + 1}{25 + 10} = \frac{9}{35} = 0.2571$$

$$P(\text{great}|-) = \frac{0 + 1}{14 + 10} = \frac{1}{24} = 0.04167$$

+ : 0.5 × 0.2571 = 0.12857 − : 0.5 × 0.04167 = 0.02083

$$P(+|\text{text}) = \frac{0.12857}{0.12857 + 0.02083} = \mathbf{0.8606}$$

+1 so a zero count does not kill the whole product

Marginalise from a full joint table

X	Y	Z	P
x1	y1	z1	.06
x1	y1	z2	.09
x1	y2	z1	.14
x1	y2	z2	.11
x2	y1	z1	.12
x2	y1	z2	.18
x2	y2	z1	.20
x2	y2	z2	.10

P(x1) = add every row with x1 (sum out Y, Z)

P(x1) = .06 + .09 + .14 + .11 = **0.40**

P(z1) = .06 + .14 + .12 + .20 = 0.52

P(x1, z1) = .06 + .14 = 0.20

$$P(x1 \mid z1) = \frac{0.20}{0.52} = \mathbf{0.3846}$$

given part goes on the bottom

Counting rows / models

rows = 2^n (n = number of binary variables)

$2^5 = \mathbf{32}$ $2^7 = \mathbf{128}$

each binary variable doubles it. 3 values each → 3^n

FILL-IN-THE-BLANK ONE-LINERS

Formulas they blank out

Greedy f(n) = **h(n)** UCS f(n) = **g(n)** A* f(n) = **g(n) + h(n)**

BFS = **FIFO queue** DFS = **LIFO stack**

IG = H(parent) − **weighted avg H(children)** pure node H = **0**, 50/50 H = **1**

σ(0) = **0.5** ReLU(neg) = **0** tanh(0) = **0**

U(s) = R(s) + γ $\mathbf{max_a}$ **Σ P(s'|s,a) U(s')** (Bellman)

VAE reparam: z = μ + **σ** · ε

P ⇒ Q ≡ **¬P ∨ Q** P ⇔ Q ≡ **(P ⇒ Q) ∧ (Q ⇒ P)**

¬(A ∧ B) ≡ **¬A ∨ ¬B** ¬(A ∨ B) ≡ **¬A ∧ ¬B**

P ⇒ Q ≡ **¬Q ⇒ ¬P** (contrapositive)

marginalise = **sum** over the other variables joint rows = $\mathbf{2^n}$

P(A|B) = **P(A ∧ B) / P(B)** normalise: α = **1 / (s(b) + s(¬b))**

category reasoning: perceive → **categorise** → predict

the bold part is what usually gets blanked